

C_ABAPD

SAP Certified Associate - Back-End Developer - ABAP Cloud

SAP CERTIFICATION TASKS

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Typographic Conventions

American English is the standard used in this handbook.

The following typographic conventions are also used.

This information is displayed in the instructor's presentation	
Demonstration	
Procedure	
Warning or Caution	
Hint	
Related or Additional Information	
Facilitated Discussion	
User interface control	<i>Example text</i>
Window title	<i>Example text</i>

Contents

Unit 1:	System-based Assessment
1	Task 1: Introduction
2	Task 2: Context
4	Task 3: Create A Package
5	Task 4: Create a Class Hierarchy
7	Task 5: Read a List of Flight Connections
9	Task 6: Create A RAP Business Object
10	Task 7: Create a CDS Access Control
11	Task 8: Add Exception Handling to a Method
12	Task 9: Create ABAP Dictionary Types

Introduction

In this assessment, you will work on one or more exercises that you must complete in the assigned system. Please make sure to read the instructions below carefully before accessing the system.

Before starting:

- Ensure your browser is up to date and clear your browser cache and cookies.
- Make sure to close all browser sessions before you start the assessment and remember not to open unnecessary tabs or windows while working on the exercises.

During the assessment:

- Ensure you follow the assessment instructions precisely. Any deviation may result in failure of the exercise.
- Avoid access to any other practical systems during the assessment in any other browser session (including incognito).
- If you accidentally close the assessment browser, go back to learning.sap.com, log in again, and go back to the assessment.
- Should you encounter technical difficulties with the assessment, please refer to the [FAQs](#) or [create a case](#) with the support team.

Ending the assessment

- Select the *Confirm completion* button to submit your assessment. This button is in the browser tab where you initially accessed the assessment, and not in the system tab.
- Once you submit the assessment, you can no longer make changes. Do not attempt to access the system again once you have submitted your assessment.
- To get the result of your assessment, please wait for the validation process to complete (approximately 15 minutes). Your result (failed/passed) will be displayed under *My Certifications*. Click on your profile icon in the top-right corner of learning.sap.com and select *My Certifications*.

Good luck with your assessment!

Context

About the Exam

In this exam you must complete practical ABAP development tasks. Please note the following points, as failure to do so may have a negative effect on your final result:

- Name all repository objects as specified in the exercise descriptions.
- If specific types for variables or method parameters are named in the exercise descriptions, you must use them.
- You must activate all repository objects that you create in order for your objects to be evaluated. Note that this does not apply to packages or message classes, which cannot be activated.

You may use any everyday aids such as learning.sap.com, help.sap.com and the ABAP Keyword Documentation (F1 Help) to help you complete the tasks. It is also expected that you will use Joule in ADT to assist you.

Logging On to the System

You will complete this exam using your own installation of Eclipse and ABAP Development Tools for Eclipse. Before starting the exam, ensure that your Eclipse installation has release 2025-12 or higher and that your ADT installation has release 3.56.0 or higher.

When you are ready to start the exam, click on *Access Practice System*. A browser window will open. You need this window to find out your group number for the exercises and the service URL for the ABAP project in ADT.



Figure 1: User Settings

To find out your group number, click on the circle containing the letters **EL**. Your group number, which you need in order to name your objects correctly during the exam, is the four-digit number following the letter **E** - so in this case, **0007**. You replace the four pound signs (####) from the task instructions with this number whenever it is used.



Figure 2: Service URL

In the address bar of the browser fenster is a URL. The highlighted part of this - up to the end of **hana.ondemand.com** - is the service URL that you will need to create your ABAP Cloud Project for the exam. Mark this URL and copy it to your clipboard. Note that it will not necessarily be identical to the URL shown, as different systems are used for the exam at different times.

Open Eclipse and create a new ABAP Cloud project using the URL that you copied from the browser window. When Eclipse prompts you to logon, use the *Copy URL to Clipboard* option and paste the logon URL into an incognito (InPrivate) browser window. This will prevent conflicts with any other logon data that you might have stored in a browser. The exam enrollment process ensures that you are logged on by single sign-on. You will not be prompted for a user and password.

**Note:**

During the exam, you will need to copy certain objects. To facilitate this, add the package **ZC_ABAPD_EXAM** to your list of favorite packages.

You have three hours to complete the exam.

Create A Package

In order to complete the subsequent exercises, you must create a package in the system.

In your examination system, create a package called **ZABAP_####** where #### is your seat number. Assign it to superpackage **ZSTUDENTS**, and create a transport request. Use this transport request in all subsequent exercises.

Make sensible entries in all mandatory fields.

Unit 1

Task 4

Create a Class Hierarchy

In your package **ZABAP_####**, create global classes according to the following specification.

You can use the definitions and contents of the database tables **/DMO/FLIGHT**, **/DMO/CONNECTION**, and **/LRN/AIRPORT** to familiarize yourself with the data.

There shall be a general class **flight** from which a specialized class **passenger_flight** is derived.

Create a global class **ZCL_####_FLIGHT** in your package **ZABAP_####**. It should have the following attributes:

Name	Type	Visibility
carrier_id	/DMO/CARRIER_ID	A consumer should be able to read the contents of the attributes but not change them
connection_id	/DMO/CONNECTION_ID	
airport_from	/DMO/AIRPORT_FROM_ID	
airport_to	/DMO/AIRPORT_TO_ID	
plane_type	/DMO/PLANE_TYPE_ID	A consumer should not be able to access the attribute, but a subclass must be able to access it directly

and the following method

Name: Constructor

Signature:

- An importing parameter for each of the attributes **carrier_id**, **connection_id**, and **plane_type**.
- The exception **ZCX_C_ABAPD_NO_CONNECTION**.

Implementation of class **ZCL_####_FLIGHT**:

1. Set the values of **carrier_id**, **connection_id**, and **plane_type**.
2. Read the corresponding entry from database table **/dmo/connection** to fill the attributes **airport_from** and **airport_to**. If the entry does not exist, raise the exception **ZCX_C_ABAPD_NO_CONNECTION**.

Create another global class **ZCL_####_PASSENGER_FLIGHT** that is derived from **ZCL_####_FLIGHT** and has the following additional attribute:

Name	Type	Visibility
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seats_max	/DMO/PLANE_SEATS_MAX	Only the class itself should have access to the attribute
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and the following method:

Name Constructor

Signature:

- An importing parameter for each of the attributes **carrier_id**, **connection_id**, and **plane_type**.
- The exception **ZCX_C_ABAPD_NO_CONNECTION**

1. Call the superclass constructor.
2. Use CDS view **ZI_CABAPD_PASSENGER** to read the field MaximumSeats for the plane type passed to the method and assign it to the seats_max attribute. If the entry does not exist, raise the exception **ZCX_C_ABAPD_NO_CONNECTION**.

Hint: If you want to test your code, you can use plane types A320-200, 737-800, or 747-400

Unit 1

Task 5

Read a List of Flight Connections

Your task is to create a method that returns a list of flight connections departing from a particular airport. The list should contain all direct flights from that airport, as well as all destinations that can be reached with a single change using the same airline. Return flights do not count.

For example, if the database contains the following entries, and the departure airport is EDI:

carrier_id	connection_id	airport_from_id	airport_to_id
LH	400	FRA	JFK
LH	2010	EDI	FRA
SQ	299	EDI	SIN
SQ	300	SIN	EDI
SQ	30	SIN	SYD
BA	447	EDI	LHR
AA	12	LHR	BOS

The result would look like this. Note that the connection EDI to EDI via SIN does not count, as it is a simple return flight, not an onward connection:

carrier_id	airport_from_id	airport_to_id	airport_via_id
LH	EDI	JFK	FRA
LH	EDI	FRA	-
SQ	EDI	SIN	-
SQ	EDI	SYD	SIN
BA	EDI	LHR	-

1. In your package **ZABAP_####**, create a class **ZCL_####_CONNECTIONS**.
2. Create a public instance method **get_connections**. It should have an importing parameter for a departure airport. Call it **i_departure** and use the type **/dmo/airport_from_id**. There should be a returning parameter for a list of connections. Call it **r_connections** and use the internal table type **zcert_connections**.
3. Solve the task using one or more **SELECT** statements that read from database table **/dmo/connection**. Note that there is more than one valid solution

4. Your target table **r_connections** should contain the values of the database fields **carrier_id**, **airport_from_id**, and **airport_to_id**. There is an additional field **airport_via_id**. If the connection is direct, place a dash '-' in the field. If the connection involves a change of flight, fill the field with the code of the airport where the passenger will have to change.

Create A RAP Business Object

You are going to copy a database table and use your copy as the basis for a generated RAP Business Object. You will add a simple field control and determination to the object.

The table contains travel information. Each row corresponds to a trip with various information such as a trip number, start and end date, and a status. In this exercise, you will work with the *Status* field to ensure that it cannot be modified by a user on the UI. Instead, you will implement a determination to automatically set the status of a newly-create trip to **N**.

1. Copy the database table **ZABAPDTRAVEL**. Call your copy **ZZ####TRAVEL** and assign it to your package **ZABAP_####**. Note that the double Z at the beginning of the name is important. Activate the table.
2. Use the *Generate ABAP Repository Objects* function from the context menu of your new table to create an *OData UI Service*. Do not change any naming proposals in the wizard.
3. In the behavior definition **ZR_Z####TRAVEL**, set the field **Status** to read-only.
4. Declare a determination **setInitialStatus** for event **Save**. It should only be executed when an instance is newly created.
5. Implement the determination so that the status of each new trip is set to **N** when the user saves the data.

Result

You can test the behavior of your UI service by publishing the service binding **ZUI_Z####TRAVEL_O4**, and choosing *Preview*.

Create a CDS Access Control

You are going to copy a CDS view, and you need to provide it with an access control.

1. Copy the CDS View Entity **ZC_ABAPD_AGENCY**. Call your copy **Z####_AGENCY** and assign it to your package **ZABAP_####**. Activate the view entity.
2. Create a CDS Access Control to provide an authorization check for the CDS View Entity. Call the access control **Z####_AGENCY**. In it, use the authorization object **/LRN/AGCY** to restrict access to the data according to the contents of the view column **agency_id**. Use the activity code corresponding to the operation *display*.

Add Exception Handling to a Method

You are going to copy a class that contains a method without any exception handling. Your task is to implement this.

1. Copy the class **ZCL_AGENCY_MODEL**. Call the copy **ZCL_####_AGENCY_MODEL** and assign it to your package **ZABAP_####**.
2. Create a global exception class **ZCX_####_NO_AGENCY**. In the class, create a constant representing message **002** from message class **ZC_ABAPD**. This message has a placeholder **&1** for the travel agency. Ensure that the agency number is passed to the exception object.
3. In the method **zcl_####_agency_model->get_agency()**, declare and raise the exception if the select statement does not return any data.

Unit 1

Task 9

Create ABAP Dictionary Types

In this task you will create a structure type and an internal table type in the ABAP Dictionary.

You require an internal table type to manage data from table **/DMO/CUSTOMER**. The data might look like this:

CUSTOMER_ID	FIRST_NAME	LAST_NAME	STREET	POSTAL_CODE	CITY	COUNTRY_CODE
000001	Theresa	Buchholm	Lerchenstr. 82	71116	Gaertringen	DE
000002	Johannes	Buchholm	Muehlalstr. 14	68723	Schwetzingen	DE
000003	James	Buchholm	Froschstr. 91	90419	Nuernberg	DE
000004	Thomas	Buchholm	Hauptstr. 84	63263	Neu-Isenburg	DE
000005	Anna	Buchholm	Hasnerstrasse 13	4020	Linz	AT

The key value is **CUSTOMER_ID**, and it is guaranteed to be unique. In the application, you will fill the table once and make repeated read accesses to the table using the key.

1. In your package **ZABAP_####**, create a structure called **Z####_CUSTOMER**, with the following components:

- **CUSTOMER_ID**
- **FIRST_NAME**
- **LAST_NAME**
- **STREET**
- **POSTAL_CODE**
- **CITY**
- **COUNTRY_CODE**

For the types of the components, use the same data elements as the corresponding fields in table **/DMO/CUSTOMER**.

Set the enhancement category of the structure to **#EXTENSIBLE_CHARACTER_NUMERIC**.

2. In your package **ZABAP_####**, create an internal table type called **Z####_T_CUST**. Use the structure that you create as its line type and make it a hashed table. Choose the correct key components based on the information above.